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# Fuzzy TOPSIS technique for multi-criteria group decision-making: A study of crude oil price

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ABSTRACT

Understanding the state of the world economy is improved by forecasting the price from oil industry. The field of crude oil price forecasting have recently heard about the technique for order preference by similarity to ideal solution (TOPSIS) and fuzzy TOPSIS (FTOPSIS) techniques; while choosing the crude oil that counteract in global oil spill reactions. A multi-criteria decision-making (MCDM) challenge has to weight several options according to various criteria. The present study, initially describes type-1 FTOPSIS technique. Secondly, it describes its extension to handle the uncertain data, known as type-1 FTOPSIS technique in multi-criteria group decision making (MCGDM). Thirdly, it also describes type-1 FTOPSIS for group decision-making (DM) to rating the response choices to a simulated crude oil price, which is one of the biggest crude oil reservoirs in the world. The outcome demonstrates the type-1 fuzzy TOPSIS framework for determining the optimal solution by considering the crude oil globally.

## 1. Introduction

In the global economy, the price of crude oil is crucial. To compose hydrocarbon deposits and other organic components, it is a naturally occurring liquid petroleum product [1]. Various techniques have been employed to choose from a limited set of options, when faced with multiple competing criteria in MCDM. The method for evaluating the performance of alternatives by comparing them to the ideal solution, is one of these FTOPSIS. The approach suggests that the optimal choice would be the one that is most like the positive ideal solution (PIS) which maximises benefit and minimises cost and least like the negative ideal solution (NIS) with maximum expense and minimum benefit [2].

When applied to a range of problems in science and engineering, the theory of fuzzy sets and fuzzy logic has proved adequate to model uncertainty or lack of understanding [3]. Fuzzy MCDM, which addresses the ambiguity, ignorance, and vagueness that are inherent in human decision-making, was first introduced by Bellman and Zadeh [4]. The decision matrix, which is formulated using criteria and alternatives, the results of developing a model for MCDM. Fuzzy number is an extension of an interval with different membership grades [5]. Fuzzy numbers behave according to their own set of laws. Many MCDM techniques have been developed in the last few decades to characterise uncertain data using fuzzy logic [6].

TOPSIS and FTOPSIS techniques are frequently used to solve decision-making problems in the real world. The subjective evaluations in the conventional TOPSIS technique are expressed as numerical values. FTOPSIS technique which has been successfully used to solve a variety of MCDM problems, has been developed to deal with MCDM with an uncertain decision matrix [7]. To utilizing the

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FTOPSIS approach, create a fuzzy technique for grouping fuzzy techniques. The work starts with a thorough review of the literature, looking at several fuzzy approach models, used in the decision-making process [8].

Developing the ideal response for battle situations is challenging since it relies on multiple variables, including the type and amount of oil spilled as well as its location. The objective is to design battle strategies for a range of circumstances that evolve over time. The application of Lagrangian and Eulerian computational models for simulation has been conducted, demonstrating encouraged outcomes [9]. In this paper, create a variety of response scenarios and choose them based on parameters like oil gathered or that reaches the coast. The choice matrix is formed in part by the evaluation of the alternatives according to these criteria. The effects of employing various battle methods for each given criteria can be assessed using simulation results. As a result, the impact type is required to provide tools for evaluating a decision's effects in any case.

Here, we propose a FTOPSIS for group decision-making and apply it to a pertinent crises management scenario to assist in choosing the most effective crude oil combat alternatives [10]. To the best of our knowledge, this technique is being used for the first time to manage a contingency plan in the event of a global crude oil disaster. Creating a tool to support decision-makers in the oil spill contingency plan which was the aim of this effort. The instrument should assist decision-makers in making tactical and operational decisions by offering the best options for an oil spill contingency plan [11].

The tariff criteria are reduced and the profit criteria are maximised in the positive optimum solution. The tariff criteria benefit the most and the profit criteria are minimised in the negative optimum solution. For ideas on real-world TOPSIS applications [12]. In case of availability of ambiguous, imprecise or uncertain information, it can be challenging to carefully consider all of the criterion's possibilities. Fuzzy numbers can be used to indicate the classification of one-by-one alternatives in relation to the one-by-one criterion [7]. An ambiguous figure arises, as one could expect from the extension of an interval with different membership assessments. In the interval, a real number is given a particularised value that perfectly captures its consistency with the ambiguous statement that includes a fuzzy number. Campaign imperative subdue fuzzy numbers. Many fuzzy logic-based MCDM techniques have emerged in recent decades to analyse unclear data.

Meanwhile, TOPSIS is enhanced to provide MCDM with a confusing decision matrix, resulting in FTOPSIS. Several MCDM issues have been solved reasonably well with FTOPSIS [3,10]. A consensus model for multi-attribute group decision-making (MAGDM) using a heterogeneous TOPSIS technique with several attribute sets, allow each decision-maker to freely select assessment criteria, assign weights to those factors and share evaluation results [7]. An anti-biassed statistical technique, aims to develop a decision support system for GDM. Using the extreme, moderate, and soft forms of this technique, decision-makers can identify and manage bias based on the relative confidence interval and overlap ratio. Expanded the TOPSIS technique using ordered fuzzy numbers to facilitate group decision-making [13].

The expansion of the fuzzy TOPSIS approach in group decision-making was explored by Krohling and Campanharo. One of the biggest oil resources in the world, uses this expanded method to assess the alternatives [10]. In this study, they presented MAGDM approach that considers opinion dynamics to minimise the information loss from the opinion dimension. An assessment of the public experience in a new smart city is utilised to show the feasibility of the proposed MAGDM approach. Then author discussed the fuzzy analytical hierarchy process (FAHP) and FTOPSIS method for determining the MCDM problems using triangular fuzzy numbers [14]. Next, it was explained how the Kandoleh dam locations in the fuzzy TOPSIS approach works. The best ranking outcomes were compared with five MCDM techniques, including TOPSIS, ELECTRE, SAW, VIKOR, and PROMETHEE, for better offshore sites [15].

Now they proposed the TOPSSIS technique for solving the MAGDM problem using linguistics values [16]. The extension of fuzzy TOPSIS is also introduced in the GDM problem under different criterion [17]. Then author discussed the generalized fuzzy TOPSIS method in MCDM problem using some linguistics terms [18]. The integrated decomposed fuzzy sets with reliability data supplied by Z-fuzzy TOPSIS are proposed, to represent both functional and dysfunctional judgements for accounting the decision makers consistency in the selection of transfer location center [19]. Subsequently, we presented the MAGDM problem, a novel decision-making technique including dual reference points. With this strategy, group decision-making behaviours should be better captured in terms of the impacts of target objectives and bottom-line concerns. They presented a novel hybrid prediction model in order to improve the accuracy of predicting prices of crude oil in sustainable development [20].

Unlike earlier research that mostly used conventional oil price series to investigate pertinent effects, the crude oil volatility index (OVX) analyses the variance of returns in alternative energy companies, in a different way [21]. A structural vector autoregressive (SVAR) model has been presented recently. Its purpose is to analyse the dynamics of the global oil market by identifying and characterising the various types of stocks, including those related to oil supply affecting global economic growth [22]. The primary task is to rank each alternative in the GDM using ordered fuzzy number by the extended TOPSIS approach [23].

To create a fuzzy model that integrates financial measures and decision-makers subjective assessments and to evaluate the performance of businesses; FAHP and FTOPSIS approaches are combined in the suggested methodology [3]. So, explore a novel approach for solving complex market challenges and chrome averaging a hybrid systematic design, serves as the primary source of information [24]. In recent developments, variational mode decomposition (VMD) has been introduced as a valuable addition to the realm of crude oil price prediction [1]. Next, introduced a novel compromise method (CR) to formulate a methodology for fuzzy MAGDM (FMAGDM) [25]. In the Bayesian model average (BMA), the spike and slab approach, and the dynamic Bayesian structural time series model (DBSTS) are used to summarize the new structural features and key determinants of crude oil prices [26].

The study of four different forecasting techniques: ARIMA (auto-regression integrated moving average), GP (genetic programming), ANN (artificial neural network), and LSSVM (least squares support vector machine) to predict the annual interest rate (AIR), daily petrol price (DGP), and monthly oil price (MOP) the meta-heuristic bat algorithm is used [27]. Using chemical dispersant and hypothetical oil spill, this study describes a numerical modelling effort to understand the possible hydrocarbon distribution in a shallow coastal bay [9]. The impact of crude oil uncertainty on aggregate and market returns across several economic sectors was

**Table 1**
Comparative studies of existing literature review.

| Authors \ Methods | TOPSIS | FTOPSIS | Min-max | Extension type-1 FTOPSIS | | GDM |
|---|---|---|---|---|---|---|
| [19] Tuysuz & Kahraman (2024) | × | ✓ | × | × | | × |
| [40] Sahoo et al. (2024) | × | ✓ | ✓ | × | × | |
| [16] Baral et al. (2023) | ✓ | × | × | × | × | |
| [14] Baral et al. (2023) | × | ✓ | × | × | × | |
| [48] Xu et al. (2023) | ✓ | × | × | × | × | |
| [8] Parida (2020) | × | ✓ | × | × | × | |
| [23] Kacprzak (2020) | ✓ | ✓ | × | × | × | |
| This paper | ✓ | ✓ | × | ✓ | ✓ | |

examined [28], utilising the importance of the crude oil price volatility index. The aim of this research is to develop a resilient multi-stage multi-criteria GDM (MSMCGDM) approach that tackles the difficulties associated with multi-stage GDM situations that are typified by a variety of information sources [29]. The study employed a STAR model that included exogenous variables to evaluate the non-linearity of WTI (West Texas Intermediate) crude oil prices [30]. The integrated fuzzy group decision-making method presented in this research addresses the inherent fuzziness in the preferences of decision makers [31]. This preference degrees of triangular fuzzy numbers are also used in MAGDM problem [32].

First, talk about the innovative TOPSIS technique for trapezoidal cubic hesitant space. Further extend the conventional trapezoidal cubic hesitant FTOPSIS method to resolve the MCDM method under the trapezoidal cubic hesitant FTOPSIS [33]. Then the empowerment of women in India are also determined in MCDM method under the basis of AHP-TOPSIS techniques [34]. This leads us to suggest a method for establishing a consensus that is based on the Louvain algorithm, social network theory and the bounded confidence (SNBC) model with interval numbers [35].

The expanded TOPSIS technique was covered in relation to prospect theory (PT) and trapezoidal intuitionistic fuzzy numbers (TrIFN) in GDM [36]. In the selection of container multimodal transport path, this trapezoidal fuzzy numbers are introduced under the basis of heterogeneous MCGDM [37]. Then an integrated method of complex heterogeneous MAGDM is proposed in the selection of photovoltaic power station [38]. In battery supplier selection, two-stage reaching process social network are presented in the MCGDM problem [39]. Then author discussed the min-max fuzzy TOPSIS approach for optimal college location selection in fuzzy MCDM problem [40]. The linguistics hesitant fuzzy are proposed in MAGDM problem for determining the enterprise resource planning selection [41]. This linguistics hesitant fuzzy are also presented in MCGDM problem for the selection of shared power bank [42]. Dual strategies reaching process model are introduced in MCGDM problem under the basis of probabilistic linguistics data for the ranking of the alternative [43].

Now first introduced the double parameters of TOPSIS method in MAGDM problem with the basis of intuitionistic Z-linguistics fuzzy variable [44]. This intuitionistic fuzzy sets are used for the GDM problem using fuzzy preference relations of best-worst method [45]. Then author introduced the interval number in MCDM problem. This interval number are also proposed in the MCGDM problem using the entropy weight method [46]. Then the extension of interval number i.e. interval type-2 trapezoidal are proposed in supplier selection for the MAGDM problem [47]. The interval-valued probabilistic linguistic q-rung orthopair fuzzy sets are also used in the applications of MAGDM problems based on two-stage TOPSIS approach [48]. Then the extension of intuitionistic fuzzy sets i.e. interval-valued intuitionistic fuzzy are also presented for the GDM problem using best-worst method with additive consistency [49]. Next, they have discussed the MCGDM method for determining the most effective LIB recycling procedure in a Fermatean fuzzy environment (FFE) [50]. Numerous conflicting criteria are used to characterise a finite number of options, and the MCDM has been widely used in this process. Out of these literature reviews, some literatures are shown in Table 1, which is related to present study.

The structure of the remaining portion of the article as follows: A basic understanding of fuzzy sets is covered in Section 2. In this section covered fuzzy numbers like triangular, trapezoidal, pentagonal, and their membership functions. Also, FTOPSIS technique, which is an expanded version of TOPSIS, is used to address the decision-matrix through uncertainty. Afterwards, we create a type-1 FTOPSIS for group decision-making in Section 3 by building on earlier research. This addresses the decision-makers preferences. Section 4 contains a case study for an oil field in the crude oil price. To demonstrate the approach's viability, simulation results are shown in Section 5. Section 6 provides the conclusion and recommendation for additional research.

## 2. Preliminaries

This section provides a quick introduction to some terms and meanings pertaining to fuzzy sets, fuzzy membership functions, triangular fuzzy numbers (TFN); trapezoidal fuzzy numbers; and pentagonal fuzzy numbers.

**Definition 2.1.** A fuzzy set $\widetilde{A}$ in $U$ is epitomized by a membership function $\mu_{\widetilde{A}}(u)$ which relates particularized point $u$ to a real number in the intermission $[0,\ 1]$ epitomizing the grade of membership of $u$ in $\widetilde{A}$.

**Definition 2.2.** A fuzzy set $\widetilde{A}$ of a domain of discourse $U$ is said to be convex if and only if $u_1 \in U$ and $u_2 \in U$, there exists:

$$\mu_{\widetilde{A}}(\alpha.u_1 + (1-\alpha).u_2) \ \geq \ min(\mu_{\widetilde{A}}(u_1),\ \mu_{\widetilde{A}}(u_2)) \tag{1}$$

where $\mu_{\widetilde{A}}$ is the membership function of the fuzzy set $\widetilde{A}$ and $\alpha \in \ [0,\ 1]$.

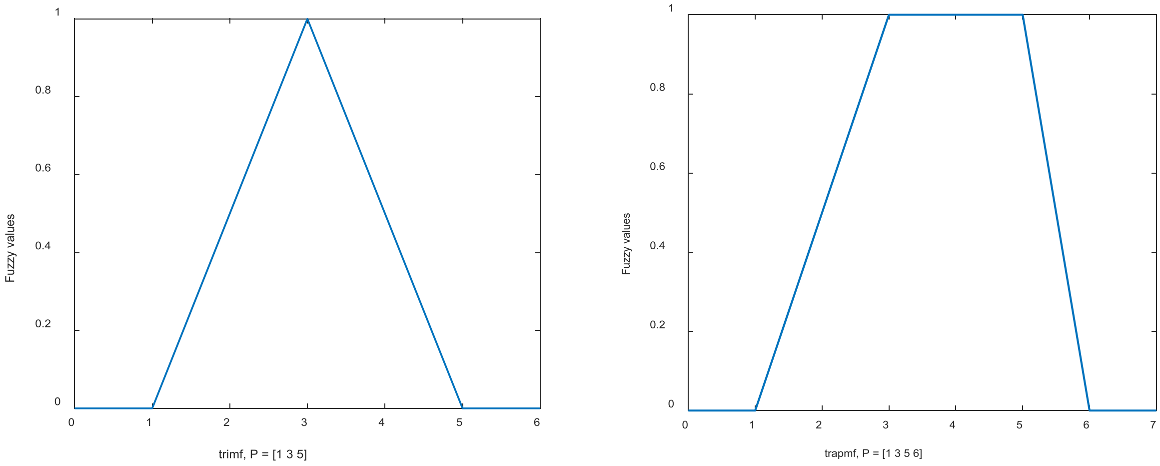



**Fig. 1.** Membership functions.

**Definition 2.3.** A triangular fuzzy number $\widetilde{a}$ is demarcated by a threeling $\widetilde{a} = (a_1,\ a_2, a_3) \in R$, (Example, Fig. 1). The membership function is written as:

$$\mu_{\tilde{a}}(u) = \begin{cases} 0, & u < a_1 \\ (u - a_1)/(a_2 - a_1), & a_1 \leq u \leq a_2 \\ (u - a_2)/(a_3 - a_2), & a_2 \leq u \leq a_3 \\ 0, & u > a_3 \end{cases} \tag{2}$$

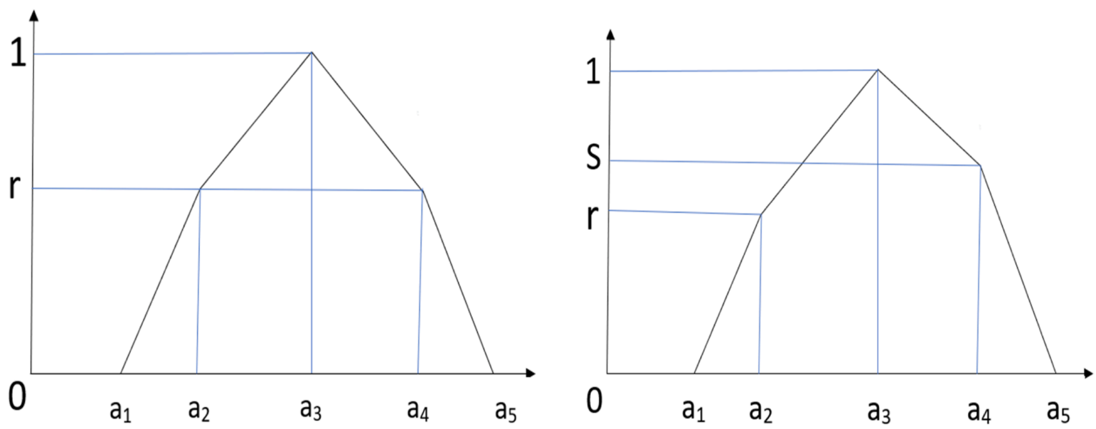
where $a_2$ epitomizes the value for which $\mu_{\tilde{a}}(a_2) = 1$, $a_1$ and $a_3$ are the uttermost appraises on the left and on the right of the fuzzy number $\widetilde{a}$ corresponding with membership $\mu_{\tilde{a}}(a_1) = \mu_{\tilde{a}}(a_3) = 0$.

**Definition 2.4.** The $\alpha$-cut of fuzzy number $\widetilde{a}$ is demarcated as

$$\widetilde{a}^{\alpha} = \{u_\beta : \mu_{\tilde{A}}(u_\beta) \geq \alpha,\ u_\beta \in U\},\ \text{where } \alpha \in [0,\ 1] \tag{3}$$

**Definition 2.5.** If $\widetilde{T}$ is a triangular fuzzy number and $[\widetilde{T}]^L_\alpha > 0,\ \ [\widetilde{T}]^U_\alpha \leq 1$ for $\alpha \in (0, 1]$ then $\widetilde{A}$ is called a normalized positive triangular fuzzy number.

**Definition 2.6.** Let $\widetilde{a} = (a_1,\ a_2, a_3) \in R$ and $\widetilde{b} = (b_1, b_2, b_3) \in R$ be two positive triangular fuzzy numbers, then the operation with these fuzzy numbers is demarcated as follows




(a) Symmetry pentagonal fuzzy number (b) Asymmetry pentagonal fuzzy number

**Fig. 2.** Pentagonal fuzzy numbers.

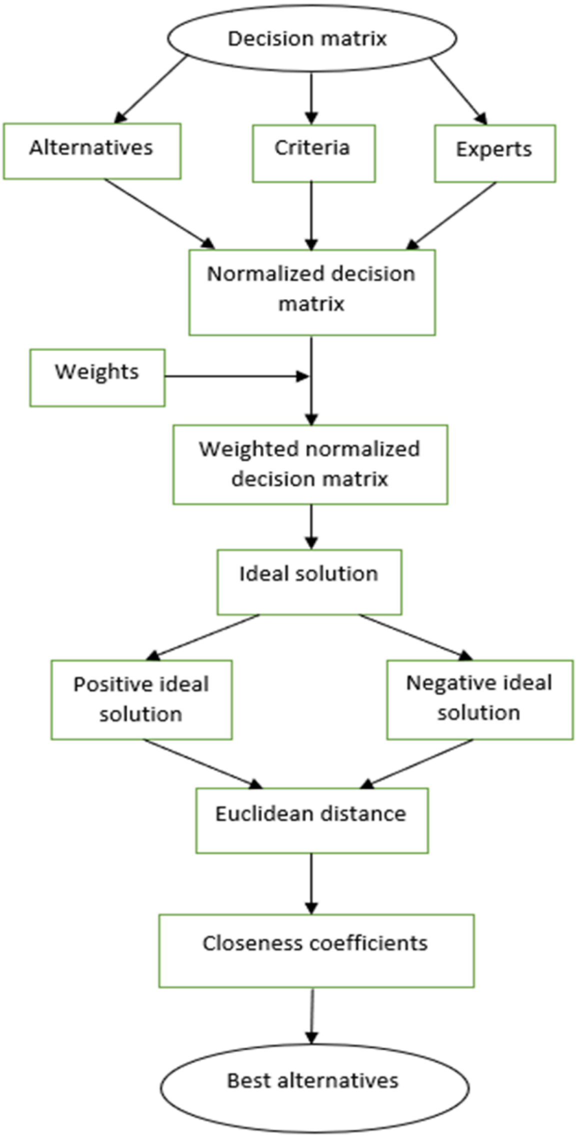



**Fig. 3.** Type-1 fuzzy TOPSIS for MCGDM.

$$\tilde{a} \oplus \tilde{b} = (a_1 \oplus b_1,\ a_2 \oplus b_2,\ a_3 \oplus b_3)$$

$$\tilde{a} \otimes \tilde{b} = (a_1 \otimes b_1,\ a_2 \otimes b_2, a_3 \otimes b_3)$$

$$\tilde{a} \ominus \tilde{b} = (a_1 \ominus b_1,\ a_2 \ominus b_2,\ a_3 \ominus b_3)$$

$$\tilde{a} \oslash \tilde{b} = (a_1 \oslash b_1, a_2 \oslash b_2, a_3 \oslash b_3)$$

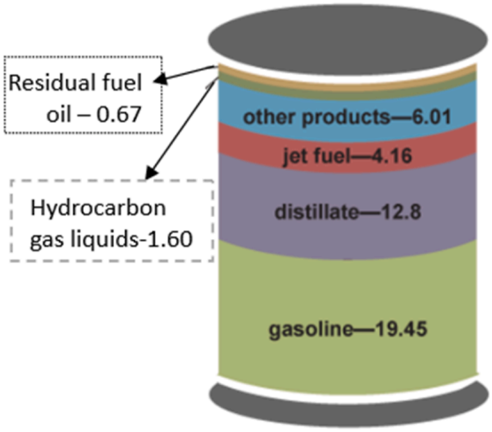
$$k\,\tilde{a} = (k\,a_1,\ k\,a_2,\ k\,a_3)$$

**Definition 2.7.** Let two triangular fuzzy numbers $\tilde{a} = (a_1,\ a_2,\ a_3) \in R$ and $\tilde{b} = (b_1,\ b_2,\ b_3) \in R$, then the interspace is computed by

$$d(\tilde{a},\ \tilde{b}) = \sqrt[2]{\frac{1}{3}\ \left[(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2\right]} \in R \tag{4}$$

**Definition 2.8.** A trapezoidal fuzzy number $T_p$, can be specified by an ordered quadruple $(a_1, a_2, a_3, a_4) \in R$ with $a_1 \le a_2 \le a_3 \le a_4$ (Example, Fig. 1) and a membership function consists of three-line segments. The membership function for a trapezoidal fuzzy number is

(a) Total stockpiles in storage facilities.




(b) Illustration: petroleum products made from a barrel of crude oil, 2022

**Fig. 4.** Petroleum stockpiles.

$$\mu_{\tilde{a}}[a_1, a_2, a_3, a_4](u) = \begin{cases} 0 & \text{if} \quad u < a_1 \\ \dfrac{u - a_1}{a_2 - a_1} & \text{if} \quad a_1 \le u \le a_2 \\ 1 & \text{if} \quad a_2 < u < a_3 \\ \dfrac{u - a_4}{a_3 - a_4} & \text{if} \quad a_3 \le u \le a_4 \\ 0 & \text{if} \quad u > a_4 \end{cases} \tag{5}$$

**Definition 2.9.** Let two trapezoidal fuzzy numbers $\widetilde{a} = (a_1, a_2, a_3, a_4) \in R$ and $\widetilde{b} = (b_1, b_2, b_3, b_4) \in R$, then the distance is computed by

$$d(\widetilde{a}, \widetilde{b}) = \sqrt[2]{\left[(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2 + (a_4 - b_4)^2\right]} \in R \tag{6}$$

**Definition 2.10.** A pentagonal fuzzy number $\widetilde{a} = (a_1, a_2, a_3, a_4, a_5) \in R$ should satisfy the following condition.

**Table 2**
Decision matrix composed of yearly COP combat alternatives and criteria.

| Alternatives | YACP ($) | YOP ($) | YHP ($) | YLP ($) | YCP ($) |
|---|---|---|---|---|---|
| $A_{2001}$ | 25.98 | 27.29 | 32.21 | 17.50 | 19.96 |
| $A_{2002}$ | 26.19 | 21.13 | 32.68 | 18.02 | 31.21 |
| $A_{2003}$ | 31.08 | 31.97 | 37.96 | 25.25 | 32.51 |
| $A_{2004}$ | 41.51 | 33.71 | 56.37 | 32.49 | 43.36 |
| $A_{2005}$ | 56.64 | 42.16 | 69.91 | 42.16 | 61.06 |
| $A_{2006}$ | 66.05 | 63.11 | 77.05 | 42.16 | 60.85 |
| $A_{2007}$ | 72.34 | 60.77 | 99.16 | 55.90 | 95.95 |
| $A_{2008}$ | 99.67 | 99.64 | 145.31 | 50.51 | 44.60 |
| $A_{2009}$ | 61.95 | 46.17 | 81.03 | 34.28 | 79.39 |
| $A_{2010}$ | 79.48 | 81.52 | 91.48 | 64.78 | 91.38 |
| $A_{2011}$ | 94.88 | 91.59 | 113.39 | 75.40 | 98.83 |
| $A_{2012}$ | 94.05 | 102.96 | 109.39 | 77.72 | 91.38 |
| $A_{2013}$ | 97.98 | 93.14 | 110.62 | 86.65 | 98.17 |
| $A_{2014}$ | 93.17 | 95.14 | 107.95 | 53.45 | 53.45 |
| $A_{2015}$ | 48.66 | 52.72 | 61.36 | 34.55 | 37.13 |
| $A_{2016}$ | 43.29 | 36.81 | 54.01 | 26.19 | 53.75 |
| $A_{2017}$ | 50.80 | 52.36 | 60.46 | 42.48 | 60.46 |
| $A_{2018}$ | 65.23 | 60.37 | 77.41 | 44.48 | 45.15 |
| $A_{2019}$ | 56.99 | 46.31 | 66.24 | 46.31 | 61.14 |
| $A_{2020}$ | 39.68 | 61.17 | 63.27 | 11.26 | 48.52 |
| $A_{2021}$ | 68.17 | 47.62 | 84.65 | 47.62 | 75.21 |
| $A_{2022}$ | 94.53 | 76.08 | 123.70 | 71.59 | 80.51 |

1. $\mu_{\tilde{a}}(u)$ is continuous function in the interval $[0,\ 1]$.
2. $\mu_{\tilde{a}}(u)$ is strictly increasing and continuous function on $[a_1, a_2]$ and $[a_2, a_3]$.
3. $\mu_{\tilde{a}}(u)$ is strictly decreasing and continuous function on $[a_3, a_4]$ and $[a_4, a_5]$.

**Definition 2.11.** Let two pentagonal fuzzy numbers $\widetilde{a} = (a_1, a_2, a_3, a_4, a_5) \in R$ and $\widetilde{b} = (b_1, b_2, b_3, b_4, b_5) \in R$, then the distance is computed by

$$d(\widetilde{a}, \widetilde{b}) = \sqrt[2]{\left[(a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2 + (a_4 - b_4)^2 + (a_5 - b_5)^2\right]} \in R \tag{7}$$

**Definition 2.12.** Two pentagonal fuzzy number $\widetilde{a} = (a_1, a_2, a_3, a_4, a_5) \in R$ and $\widetilde{b} = (b_1, b_2, b_3, b_4, b_5) \in R$ are said to be equal if $a_1 = b_1,\ a_2 = b_2,\ a_3 = b_3,\ a_4 = b_4,\ a_5 = b_5$.

**Definition 2.13.** A linear pentagonal fuzzy number with symmetry is written as $\widetilde{a}_{LS}([a_1, a_2, a_3, a_4, a_5];\ r)$, (Example, Fig. 2) whose membership function is written as

$$\mu_{\tilde{a}_{LS}}[a_1, a_2, a_3, a_4, a_5](u) = \begin{cases} 0 & if\ \ u < a_1 \\ r\dfrac{u - a_1}{a_2 - a_1} & if\ \ a_1 \le u \le a_2 \\ 1 - (1 - r)\dfrac{u - a_2}{a_3 - a_2} & if\ \ a_2 \le u \le a_3 \\ 1 & if\ \ u = a_3 \\ 1 - (1 - r)\dfrac{a_4 - u}{a_4 - a_3} & if\ \ a_3 \le u \le a_4 \\ r\dfrac{a_5 - u}{a_5 - a_4} & if\ \ a_4 \le u \le a_5 \\ 0 & if\ \ u > a_5 \end{cases} \tag{8}$$

**Definition 2.14.** A linear pentagonal fuzzy number with asymmetry is written as $\widetilde{a}_{LAS}([a_1, a_2, a_3, a_4, a_5];\ r;\ s)$, (Example, Fig. 2) whose membership function is written as

$$\mu_{\tilde{a}_{LAS}}[a_1, a_2, a_3, a_4, a_5](u) = \begin{cases} 0 & if\ \ u < a_1 \\ r\dfrac{u - a_1}{a_2 - a_1} & if\ \ a_1 \le u \le a_2 \\ 1 - (1 - r)\dfrac{u - a_2}{a_3 - a_2} & if\ \ a_2 \le u \le a_3 \\ 1 & if\ \ u = a_3 \\ 1 - (1 - s)\dfrac{a_4 - u}{a_4 - a_3} & if\ \ a_3 \le u \le a_4 \\ s\dfrac{a_5 - u}{a_5 - a_4} & if\ \ a_4 \le u \le a_5 \\ 0 & if\ \ u > a_5 \end{cases} \tag{9}$$

## 3. Multi-criteria decision-making

In this section, we describe the details about TOPSIS technique, type-1 fuzzy TOPSIS technique and its algorithms. Also describe the proposed methodology for group decision-making.

### 3.1. TOPSIS technique

Choosing decision matrix $D_T$ for an MCDM problem that has $m$ possibilities and $n$ criteria can be expressed in matrix form as $D_T = (x_{\theta\sigma})_{m\times n}$, where $A_\theta$, $\theta = 1, 2, \cdots, m$ are alternatives and $C_\sigma$, $\ \sigma = 1, 2, \cdots, n$ are the criterion. In terms of criterion $C_\sigma$, the evaluation of the alternatives $A_\theta$ is demonstrated by $x_{\theta\sigma}$ prototype scores. The weight vector $w = (w_1, w_2, \cdots, w_n)$ is unflappable of the particularized weights for $w_\sigma$, $\sigma = 1, 2, \cdots, n$ each criterion $C_\sigma$ satisfying $\sum_{\sigma=1}^{n} w_\sigma = 1$.

**Algorithm 1.**

Step 1.Assemble normalized decision matrix $N_{\theta\sigma}$, where $N_{\theta\sigma} = x_{\theta\sigma}/\sqrt{\sum x_{\theta\sigma}^2}$ for $\theta = 1, 2, \cdots, m;\ \ \sigma = 1, 2, \cdots, n$, where $x_{\theta\sigma}$ and $N_{\theta\sigma}$ are

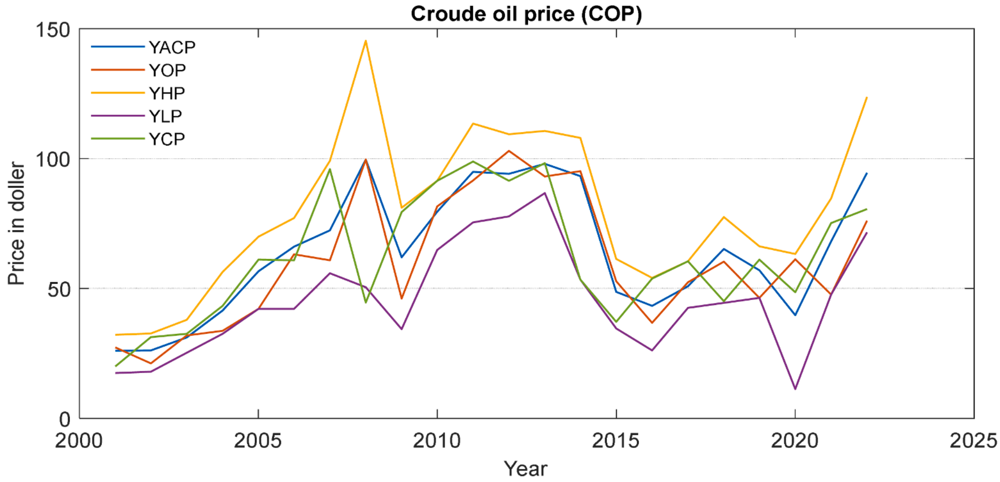



**Fig. 5.** Five criterion of crude oil price ($).

prototype and normalized score of decision matrix.

Step 2. The weighted normalized decision matrix: $V_{\theta\sigma} = w_\sigma N_{\theta\sigma}$, where $w_\sigma$ is the weight for $\sigma^{th}$ criteria and $\sum w_\sigma = 1$, where $\sigma = 1, 2, \cdots, n$.

Step 3. Determine the positive ideal solution $p^+$:

$$p^+ = (v_1^+, v_2^+, \cdots, v_n^+) = \left\{ \underset{\theta}{max} V_{\theta\sigma} \middle| \sigma \in J_1;\ \underset{\theta}{min} V_{\theta\sigma} \middle| \sigma \in J_2 \right\},$$

and the negative ideal solution $p^-$:

$$p^- = (v_1^-, v_2^-, \cdots, v_n^-) = \left\{ \underset{\theta}{min} V_{\theta\sigma} \middle| \sigma \in J_1;\ \underset{\theta}{max} V_{\theta\sigma} \middle| \sigma \in J_2 \right\},$$

where $v_\sigma^+ = (1,1,1)$; $v_\sigma^- = (0,0,0)$ and $J_1$ and $J_2$ epitomizes the benefit criteria and tariff criteria correspondingly.

Step 4. Compute the Euclidean interspaces from the positive ideal $I^+$ and negative ideal $I^-$ solutions for each alternative $A_\theta$ correspondingly:

$$d_\theta^+ = \sqrt{\sum_\sigma (\Delta_{\theta\sigma}^+)^2} \text{ and } d_\theta^- = \sqrt{\sum_\sigma (\Delta_{\theta\sigma}^-)^2},$$

where $\Delta_{\theta\sigma}^+ = (v_\sigma^+ - V_{\theta\sigma})$ and $\Delta_{\theta\sigma}^- = (v_\sigma^- - V_{\theta\sigma})$ with $\theta = 1, 2, \cdots, m$.

Step 5. The relative closeness $\Omega_\theta$ for each alternative $A_\theta$ with respect to positive ideal solution $p^+$ as given by

$$\Omega_\theta = \frac{d_\theta^-}{(d_\theta^- + d_\theta^+)},$$

where $\theta = 1, 2, \cdots, m$ and $\sigma = 1, 2, \cdots, n$.

### 3.2. Type-1 fuzzy TOPSIS technique

The fuzzy decision matrix which resides of alternatives and criteria is characterized by $D_M = (\widetilde{X}_{\theta\sigma})_{m\times n}$, where $A_1,\ A_2,\ A_3,\ \cdots, A_m$ are alternatives and $C_1,\ C_2,\ C_3, \cdots,\ C_n$ are criteria, $\widetilde{X}_{\theta\sigma}$ are fuzzy numbers demonstrates the appraisal of the alternatives $A_\theta$ with respect to criteria $C_\sigma$. The weight vector $W = (W_1, W_2,\ W_3, \cdots, W_n)$ unflappable of the particularized weights $W$ for criteria $C_\sigma$ satisfying $\sum_{\sigma=1}^n W_\sigma = 1$. The weighted normalized decision matrix $\widetilde{W}_N = [\widetilde{w}_{\theta\sigma}]_{m\times n}$ with $\theta = 1, 2, 3, \cdots, m$ and $\sigma = 1, 2, 3, \cdots, n$ assembled by multiplying the normalized fuzzy decision matrix by its concomitant weights. The weighted fuzzy normalized value $\widetilde{N}_{\theta\sigma}$ is computed as $\widetilde{N}_{\theta\sigma} = W_\sigma \times \widetilde{w}_{\theta\sigma}$ with $\theta = 1, 2, 3, \cdots, m$ and $\sigma = 1, 2, 3, \cdots, n$. The type-1 fuzzy TOPSIS method is comprise form of algorithm.

**Algorithm 2.**

Step 1. Compute the $I^+$ (positive ideal solution) and $I^-$ (negative ideal solution) as

$$I^+ = (\widetilde{N}_1^+,\ \widetilde{N}_2^+,\ \widetilde{N}_3^+, \cdots, \widetilde{N}_m^+) \text{ and } I^- = (\widetilde{N}_1^-,\ \widetilde{N}_2^-,\ \widetilde{N}_3^-, \cdots, \widetilde{N}_n^-),$$

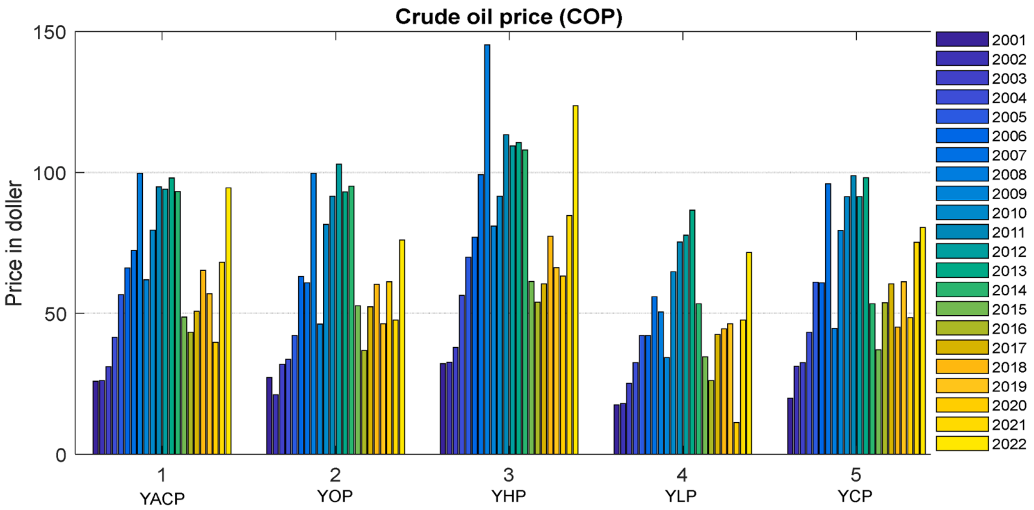



**Fig. 6.** Crude oil prices ($) of YACP, YOP, YHP, YLP, and YCP.

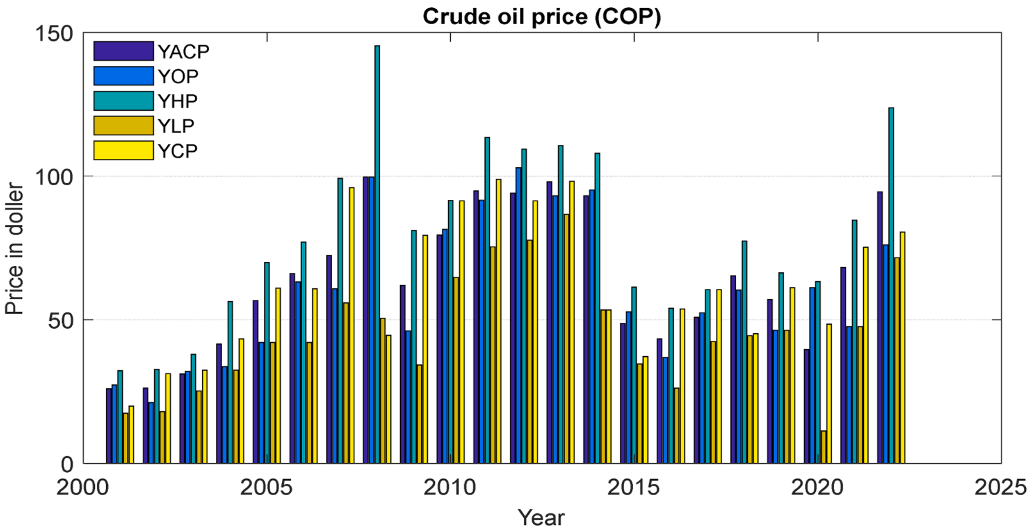



**Fig. 7.** Yearly crude oil price ($).

where $\widetilde{N}_\sigma^+ = \left(\underset{\theta}{max}\widetilde{N}_{\theta\sigma}, \sigma \in J_1;\ \underset{\theta}{min}\widetilde{N}_{\theta\sigma},\ \sigma \in J_2\right)$ and $\widetilde{N}_\sigma^- = \left(\underset{\theta}{min}\widetilde{N}_{\theta\sigma},\ \sigma \in J_1;\ \underset{\theta}{max}\widetilde{N}_{\theta\sigma},\ \sigma \in J_2\right)$, where $J_1$ and $J_2$ stands for the criteria benefits and criteria cost respectively.

Step 2. Compute the Euclidean distance between $I^+$ and $I^-$ for each alternative $A_\theta$ as follows.

$\widetilde{E}_\theta^+ = \sum_{\sigma=1}^n E(\widetilde{N}_{\theta\sigma},\ \widetilde{N}_\sigma^+)$ with $\theta = 1, 2, 3, \cdots, m$ and $\widetilde{E}_\theta^- = \sum_{\sigma=1}^n E(\widetilde{N}_{\theta\sigma},\ \widetilde{N}_\sigma^-)$ with $\theta = 1, 2, 3, \cdots, m$, where the distance $E(\widetilde{N}_{\theta\sigma},\ \widetilde{N}_\sigma^+)$ between two fuzzy numbers.

Step 3. The relative closeness coefficients $\widetilde{\Re}_\theta$ for each alternative $A_\theta$ with respect to positive ideal solution is

$\widetilde{\Re}_\theta = \frac{\widetilde{E_\theta^-}}{\widetilde{E_\theta^-} + \widetilde{E_\theta^+}}$, where $\theta = 1, 2, 3, \cdots, m$.

### 3.3. *Proposed methodology: multi-criteria group decision-making*

According to Zhang and Lu [31], the GDM framework incorporates the following features: decision-makers can assign varying weights to group members, they can convey ambiguous preferences for alternative solutions, they can make differing assessments of selection criteria, and each DM gives numbers. After weighting each member's choice for an alternative and evaluating it against the selection criteria, the group comes to a final conclusion. By adding the ideas of group PIS and NIS, Baral et al. [14] offered an alternative strategy that applies the TOPSIS framework of MCGDM scenarios.

**Table 3**
Normalized fuzzy decision matrix using triangular fuzzy numbers.

| $Al. \setminus Cr.$ | $C_{YACP}$ | $C_{YOP}$ | $C_{YHP}$ | $C_{YLP}$ | $C_{YCP}$ |
|---|---|---|---|---|---|
| $A_{2001}$ | (0.1471,0.1634, 0.1797) | (0.1545,0.1716, 0.1888) | (0.1823,0.2026, 0.2228) | (0.0991,0.1101, 0.1211) | (0.1130,0.1255, 0.1381) |
| $A_{2002}$ | (0.1482,0.1647, 0.1812) | (0.1196,0.1329, 0.1462) | (0.1850,0.2055, 0.2261) | (0.1020,0.1133, 0.1247) | (0.1767,0.1963, 0.2159) |
| $A_{2003}$ | (0.1759,0.1955, 0.2150) | (0.1810,0.2011, 0.2212) | (0.2149,0.2387, 0.2626) | (0.1429,0.1588, 0.1747) | (0.1840,0.2045, 0.2249) |
| $A_{2004}$ | (0.2350,0.2611, 0.2872) | (0.1908,0.2120, 0.2332) | (0.3191,0.3545, 0.3900) | (0.1839,0.2043, 0.2248) | (0.2454,0.2727, 0.3000) |
| $A_{2005}$ | (0.3206,0.3562, 0.3918) | (0.2386,0.2652, 0.2917) | (0.3957,0.4397, 0.4837) | (0.2386,0.2652, 0.2917) | (0.3456,0.3840, 0.4224) |
| $A_{2006}$ | (0.3739,0.4154, 0.4569) | (0,3572,0.3969, 0.4366) | (0.4361,0.4846, 0.5331) | (0.2386,0.2652, 0.2917) | (0.3444,0.3827, 0.4210) |
| $A_{2007}$ | (0.4095,0.4550, 0.5005) | (0.3440,0.3822, 0.4204) | (0.5613,0.6236, 0.6860) | (0.3164,0.3516, 0.3867) | (0.5431,0.6035, 0.6638) |
| $A_{2008}$ | (0.5642,0.6269, 0.6895) | (0.5640,0.6267, 0.6893) | (0.8225,0.9139, 1.0053) | (0.2859,0.3177, 0.3494) | (0.2525,0.2805, 0.3086) |
| $A_{2009}$ | (0.3507,0.3896, 0.4286) | (0.2613,0.2904, 0.3194) | (0.4587,0.5096, 0.5606) | (0.1940,0.2156, 0.2372) | (0.4494,0.4993, 0.5492) |
| $A_{2010}$ | (0.4499,0.4999, 0.5499) | (0.4614,0.5127, 0.5640) | (0.5178,0.5753, 0.6329) | (0.3667,0.4074, 0.4482) | (0.5172,0.5747, 0.6322) |
| $A_{2011}$ | (0.5371,0.5967, 0.6564) | (0.5184,0.5760, 0.6336) | (0.6418,0.7131, 0.7845) | (0.4268,0.4742, 0.5216) | (0.5594,0.6216, 0.6837) |
| $A_{2012}$ | (0.5324,0.5915, 0.6507) | (0.5828,0.6475, 0.7123) | (0.6192,0.6880, 0.7568) | (0.4399,0.4888, 0.5377) | (0.5172,0.5747, 0.6322) |
| $A_{2013}$ | (0.5546,0.6162, 0.6778) | (0.5272,0.5858, 0.6444) | (0.6262,0.6957, 0.7653) | (0.4905,0.5450, 0.5995) | (0.5557,0.6174, 0.6792) |
| $A_{2014}$ | (0.5274,0.5860, 0.6446) | (0.5385,0.5984, 0.6582) | (0.6110,0.6789, 0.7468) | (0.3025,0.3362, 0.3698) | (0.3025,0.3362, 0.3698) |
| $A_{2015}$ | (0.2754,0.3060, 0.3366) | (0.2984,0.3316, 0.3647) | (0.3473,0.3859, 0.4245) | (0.1956,0.2173, 0.2390) | (0.2102,0.2335, 0.2569) |
| $A_{2016}$ | (0.2450,0.2723, 0.2995) | (0.2084,0.2315, 0.2547) | (0.3057,0.3397, 0.3737) | (0.1482,0.1647, 0.1812) | (0.3042,0.3381, 0.3719) |
| $A_{2017}$ | (0.2875,0.3195, 0.3514) | (0.2964,0.3293, 0.3622) | (0.3422,0.3803, 0.4183) | (0.2405,0.2672, 0.2939) | (0.3422,0.3803, 0.4183) |
| $A_{2018}$ | (0.3692,0.4103, 0.4513) | (0.3417,0.3797, 0.4177) | (0.4382,0.4869, 0.5355) | (0.2518,0.2797, 0.3077) | (0.2556,0.2840, 0.3124) |
| $A_{2019}$ | (0.3226,0.3584, 0.3943) | (0.2621,0.2913, 0.3204) | (0.3749,0.4166, 0.4583) | (0.2621,0.2913, 0.3204) | (0.3461,0.3845, 0.4230) |
| $A_{2020}$ | (0.2246,0.2496, 0.2745) | (0.3462,0.3847, 0.4232) | (0.3581,0.3979, 0.4377) | (0.0637,0.0708, 0.0779) | (0.2746,0.3052, 0.3357) |
| $A_{2021}$ | (0.3859,0.4287, 0.4716) | (0.2695,0.2995, 0.3294) | (0.4792,0.5324, 0.5856) | (0.2695,0.2995, 0.3294) | (0.4257,0.4730, 0.5203) |
| $A_{2022}$ | (0.5351,0.5945, 0.6540) | (0.4306,0.4785, 0.5263) | (0.7002,0.7780, 0.8558) | (0.4052,0.4503, 0.4953) | (0.4557,0.5064, 0.5570) |

By simultaneously computing the distances to the group PIS and NIS, their method defines a relative closeness to the ranking order of each alternative. However, the subjective evaluations are expressed as numerical values in the conventional TOPSIS formulation, the paradigm put forward by [14] might have drawbacks. The aim is to address the decision-matrix's uncertainty while building upon the paradigm. For group decision-making, we create a novel framework that combines the prior type-1 fuzzy TOPSIS with TOPSIS in this context, as shown in Fig. 3.

With respect to individual preferences, we have already examined TOPSIS and type-1 FTOPSIS. Now, there is a group decision-makers, one of the changes in GDM. Therefore, as follows:

$$G = \{M_1, M_2, \cdots, M_K\}.$$

The following weight vectors apply to each member of the $K$ decision-makers in the group:

$W^k = (w_1^k, w_2^k, \cdots, w_n^k)$ with $k = 1, 2, \cdots, K$.where each $w_\sigma^k$ represents as the assigned to criteria $C_\sigma$ by the group member $M_k$. It is satisfying $0 \le w_\sigma^k \le 1$, and $\sum_{\sigma=1}^{n} w_\sigma^k = 1$. Also each DM has the degree of importance weights written as $0 \le \alpha_k \le 1, \ \sum_{k=1}^{K} \alpha_k = 1$.

However, we apply type-1 fuzzy TOPSIS to every group member. So, firstly we determine the fuzzy normalized decision matrix $\widetilde{I} = (\widetilde{I}_{\theta\sigma})_{m\times n}$ with $\theta = 1, 2, \cdots, m$ and $\sigma = 1, 2, \cdots, n$ for each DM, $k = 1, 2, \cdots, K$ with weight vector $W^k$. Hence, the weighted fuzzy normalized decision matrix for each DM ${}^k\widetilde{N} = ({}^k\widetilde{N}_{\theta\sigma})_{m\times n}$ is calculated as:

$${}^k\widetilde{N} = \begin{pmatrix} w_1^k\widetilde{I}_{11} & w_2^k\widetilde{I}_{12} & \cdots & w_n^k\widetilde{I}_{1n} \\ w_1^k\widetilde{I}_{21} & w_2^k\widetilde{I}_{22} & \cdots & w_n^k\widetilde{I}_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ w_1^k\widetilde{I}_{m1} & w_2^k\widetilde{I}_{m2} & \cdots & w_n^k\widetilde{I}_{mn} \end{pmatrix} \text{ with } \theta = 1, 2, \cdots, m; \ \sigma = 1, 2, \cdots, n; \text{ and } k = 1, 2, \cdots, K.$$

The steps for creating the group of alternatives are given below form of algorithm:

**Algorithm 3.**

Step 1. The $I^+$ (benefits) and $I^-$(cost) for each group member $k = 1, 2, 3, \cdots, K$ as follows.

${}^kI^+ = ({}^k\widetilde{N}_1^+, {}^k\widetilde{N}_2^+, {}^k\widetilde{N}_3^+, \cdots, {}^k\widetilde{N}_m^+)$ and ${}^kI^- = ({}^k\widetilde{N}_1^-, {}^k\widetilde{N}_2^-, {}^k\widetilde{N}_3^-, \cdots, {}^k\widetilde{N}_n^-)$where ${}^k\widetilde{N}_\sigma^+ = \left(\max_\theta {}^k\widetilde{N}_{\theta\sigma}, \sigma \in J_1; \ \min_\theta {}^k\widetilde{N}_{\theta\sigma}, \ \sigma \in J_2\right)$ and ${}^k\widetilde{N}_\sigma^- = \left(\min_\theta {}^k\widetilde{N}_{\theta\sigma}, \ \sigma \in J_1; \ \max_\theta {}^k\widetilde{N}_{\theta\sigma}, \ \sigma \in J_2\right)$, where $J_1$ and $J_2$ stands for the criteria, benefit and cost respectively.

Step 2. Compute the distance of each alternative for various members. The distance of alternatives $A_\theta$ between $I^+$ and $I^-$ of the group members $M_k$,${}^kE_\theta^+$ and ${}^kE_\theta^-$ are given as:

${}^kE_\theta^+ = \sum_{\sigma=1}^{n} E({}^k\widetilde{N}_{\theta\sigma}, \ {}^k\widetilde{N}_\sigma^+)$ and ${}^kE_\theta^- = \sum_{\sigma=1}^{n} E({}^k\widetilde{N}_{\theta\sigma}, \ {}^k\widetilde{N}_\sigma^-)$ with $\theta = 1,2,3,\cdots,m$; $k = 1,2,3,\cdots,K$, where the distance $E({}^k\widetilde{N}_{\theta\sigma}, \ {}^k\widetilde{N}_\sigma^+)$ and $E({}^k\widetilde{N}_{\theta\sigma}, \ {}^k\widetilde{N}_\sigma^-)$ between two fuzzy numbers are computed.

Step 3. The relative closeness coefficients for each alternative $A_\theta$ for each member $k$, $R(A_\theta)^k$ with respect to ${}^kE_\theta^+$ and ${}^kE_\theta^-$ as

**Table 4**
Application of type-1 fuzzy TOPSIS for $DM^1$ with weights $W_1 = (0.20, 0.20, 0.20, 0.20, 0.20)$.

| Al. \ Cr. | $W_1C_{YACP}$ | $W_1C_{YOP}$ | $W_1C_{YHP}$ | $W_1C_{YLP}$ | $W_1C_{YCP}$ | $1_{E_\theta^+}$ | $1_{E_\theta^-}$ | $R_\theta^1$ |
|---|---|---|---|---|---|---|---|---|
| $A_{2001}$ | (0.0265,0.0327, 0.0395) | (0.0278,0.0343, 0.0415) | (0.0328,0.0405, 0.0490) | (0.0178,0.0220, 0.0266) | (0.0203,0.0251, 0.0304) | 0.1315 | 0.2009 | 0.6044 |
| $A_{2002}$ | (0.0267,0.0329, 0.0399) | (0.0215,0.0266, 0.0322) | (0.0333,0.0411, 0.0497) | (0.0184,0.0227, 0.0274) | (0.0318,0.0393, 0.0475) | 0.1378 | 0.1933 | 0.5839 |
| $A_{2003}$ | (0.0317,0.0391, 0.0473) | (0.0326,0.0402, 0.0487) | (0.0387,0.0477, 0.0578) | (0.0257,0.0318, 0.0331) | (0.0331,0.0409, 0.0495) | 0.1219 | 0.1856 | 0.6037 |
| $A_{2004}$ | (0.0423,0.0522, 0.0632) | (0.0343,0.0424, 0.0513) | (0.0574,0.0709, 0.0858) | (0.0331,0.0409, 0.0495) | (0.0442,0.0545, 0.0660) | 0.1224 | 0.1570 | 0.5620 |
| $A_{2005}$ | (0.0577,0.0712, 0.0862) | (0.0430,0.0530, 0.0642) | (0.0712,0.0879, 0.1064) | (0.0430,0.0530, 0.0642) | (0.0622,0.0768, 0.0929) | 0.1265 | 0.1306 | 0.5079 |
| $A_{2006}$ | (0.0673,0.0831, 0.1005) | (0.0643,0.0794, 0.0961) | (0.0785,0.0969, 0.1173) | (0.0430,0.0530, 0.0642) | (0.0620,0.0765, 0.0926) | 0.1207 | 0.1280 | 0.5146 |
| $A_{2007}$ | (0.0737,0.0910, 0.1101) | (0.0619,0.0764, 0.0925) | (0.1010,0.1247, 0.1509) | (0.0570,0.0703, 0.0851) | (0.0978,0.1207, 0.1460) | 0.1578 | 0.1030 | 0.3950 |
| $A_{2008}$ | (0.1016,0.1254, 0.1517) | (0.1015,0.1253, 0.1517) | (0.1481,0.1828, 0.2212) | (0.0515,0.0635, 0.0769) | (0.0454,0.0561, 0.0679) | 0.1821 | 0.1324 | 0.4209 |
| $A_{2009}$ | (0.0631,0.0779, 0.0943) | (0.0470,0.0581, 0.0703) | (0.0826,0.1019, 0.1233) | (0.0349,0.0431, 0.0522) | (0.0809,0.0999, 0.1208) | 0.1473 | 0.1080 | 0.4231 |
| $A_{2010}$ | (0.0810,0.1000, 0.1210) | (0.0831,0.1025, 0.1241) | (0.0932,0.1151, 0.1392) | (0.0660,0.0815, 0.0986) | (0.0931,0.1149, 0.1391) | 0.1429 | 0.1275 | 0.4714 |
| $A_{2011}$ | (0.0967,0.1193, 0.1444) | (0.0933,0.1152, 0.1394) | (0.1155,0.1426, 0.1726) | (0.0768,0.0948, 0.1148) | (0.1007,0.1243, 0.1504) | 0.1712 | 0.1291 | 0.4298 |
| $A_{2012}$ | (0.0958,0.1183, 0.1431) | (0.1049,0.1295, 0.1567) | (0.1115,0.1376, 0.1665) | (0.0792,0.0978, 0.1183) | (0.0931,0.1149, 0.1391) | 0.1611 | 0.1434 | 0.4709 |
| $A_{2013}$ | (0.0998,0.1232, 0.1491) | (0.0949,0.1172, 0.1418) | (0.1127,0.1391, 0.1684) | (0.0883,0.1090, 0.1319) | (0.1000,0.1235, 0.1494) | 0.1699 | 0.1410 | 0.4534 |
| $A_{2014}$ | (0.0949,0.1172, 0.1418) | (0.0969,0.1197, 0.1448) | (0.1100,0.1358, 0.1643) | (0.0545,0.0672, 0.0814) | (0.0545,0.0672, 0.0814) | 0.1436 | 0.1330 | 0.4809 |
| $A_{2015}$ | (0.0496,0.0612, 0.0741) | (0.0537,0.0663, 0.0802) | (0.0625,0.0772, 0.0934) | (0.0352,0.0435, 0.0526) | (0.0378,0.0467, 0.0565) | 0.1065 | 0.1571 | 0.5959 |
| $A_{2016}$ | (0.0441,0.0545, 0.0659) | (0.0375,0.0463, 0.0560) | (0.0550,0.0679, 0.0822) | (0.0267,0.0329, 0.0399) | (0.0548,0.0676, 0.0818) | 0.1279 | 0.1519 | 0.5427 |
| $A_{2017}$ | (0.0518,0.0639, 0.0773) | (0.0533,0.0659, 0.0797) | (0.0616,0.0761, 0.0920) | (0.0433,0.0534, 0.0647) | (0.0616,0.0761, 0.0920) | 0.1116 | 0.1463 | 0.5674 |
| $A_{2018}$ | (0.0665,0.0821, 0.0993) | (0.0615,0.0759, 0.0919) | (0.0789,0.0974, 0.1178) | (0.0453,0.0559, 0.0677) | (0.0460,0.0568, 0.0687) | 0.1134 | 0.1365 | 0.5463 |
| $A_{2019}$ | (0.0581,0.0717, 0.0867) | (0.0472,0.0583, 0.0705) | (0.0675,0.0833, 0.1008) | (0.0472,0.0583, 0.0705) | (0.0623,0.0769, 0.0931) | 0.1193 | 0.1367 | 0.5340 |
| $A_{2020}$ | (0.0404,0.0499, 0.0604) | (0.0623,0.0769, 0.0931) | (0.0645,0.0796, 0.0963) | (0.0115,0.0142, 0.0171) | (0.0494,0.0610, 0.0738) | 0.1244 | 0.1543 | 0.5537 |
| $A_{2021}$ | (0.0695,0.0857, 0.1038) | (0.0485,0.0599, 0.0725) | (0.0862,0.1065, 0.1288) | (0.0485,0.0599, 0.0725) | (0.0766,0.0946, 0.1145) | 0.1415 | 0.1093 | 0.4357 |
| $A_{2022}$ | (0.0963,0.1189, 0.1439) | (0.0775,0.0957, 0.1158) | (0.1260,0.1556, 0.1883) | (0.0729,0.0901, 0.1090) | (0.0820,0.1013, 0.1225) | 0.1706 | 0.1110 | 0.3942 |
| $I^+$ | (0.0265,0.0327, 0.0395) | (0.1049,0.1295, 0.1567) | (0.0328,0.0405, 0.0490) | (0.0883,0.1090, 0.1319) | (0.0203,0.0251, 0.0304) | | | |
| $I^-$ | (0.1016,0.1254, 0.1517) | (0.0215,0.0266, 0.0322) | (0.1481,0.1828, 0.2212) | (0.0115,0.0142, 0.0171) | (0.1007,0.1243, 0.1504) | | | |

$R(A_\theta)^k = \frac{{}^kE_\theta^-}{{}^kE_\theta^+ + {}^kE_\theta^-}$ with $\theta = 1, 2, 3, \cdots, m$; $k = 1, 2, 3, \cdots, K$.

After evaluating the $R(A_\theta)^k$ for each member $k$, form the relative closeness matrix as following:

$$T = \begin{pmatrix} R(A_1)^1 & \cdots & R(A_1)^K \\ \vdots & \ddots & \vdots \\ R(A_m)^1 & \cdots & R(A_m)^K \end{pmatrix}$$

Now obtain the weighted relative closeness matrix, by introducing the degree of important weights of group members into the relative closeness is given by

$$T\alpha = \begin{pmatrix} \propto_1 R(A_1)^1 & \cdots & \propto_K R(A_1)^K \\ \vdots & \ddots & \vdots \\ \propto_1 R(A_m)^1 & \cdots & \propto_K R(A_m)^K \end{pmatrix}$$

Step 4. Identify the groups, $I^+$ and $I^-$:

$$I_G^+ = (\widetilde{N}_{G1}^+, \widetilde{N}_{G2}^+, \cdots, \widetilde{N}_{GK}^+) = \left( \max_\theta \propto_1 R(A_\theta)^1, \max_\theta \propto_2 R(A_\theta)^2, \cdots, \max_\theta \propto_K R(A_\theta)^K \right);$$

**Table 5**
The group relative closeness matrix.

| $Al. \setminus Cr.$ | $R_\theta^1$ | $R_\theta^2$ | $R_\theta^3$ | $R_\theta^4$ | $R_\theta^5$ |
|---|---|---|---|---|---|
| $A_{2001}$ | 0.6044 | 0.8832 | 0.7572 | 0.8778 | 0.7617 |
| $A_{2002}$ | 0.5839 | 0.8785 | 0.7458 | 0.8656 | 0.7554 |
| $A_{2003}$ | 0.6037 | 0.8763 | 0.7592 | 0.8666 | 0.7667 |
| $A_{2004}$ | 0.5620 | 0.7638 | 0.6901 | 0.7554 | 0.7000 |
| $A_{2005}$ | 0.5079 | 0.6465 | 0.6176 | 0.6393 | 0.6262 |
| $A_{2006}$ | 0.5146 | 0.5858 | 0.5869 | 0.5872 | 0.5841 |
| $A_{2007}$ | 0.3950 | 0.3972 | 0.4150 | 0.3879 | 0.4214 |
| $A_{2008}$ | 0.4209 | 0.1437 | 0.2363 | 0.1896 | 0.2004 |
| $A_{2009}$ | 0.4231 | 0.5463 | 0.5333 | 0.5375 | 0.5379 |
| $A_{2010}$ | 0.4714 | 0.4556 | 0.4962 | 0.4513 | 0.4968 |
| $A_{2011}$ | 0.4298 | 0.2833 | 0.3598 | 0.2839 | 0.3573 |
| $A_{2012}$ | 0.4709 | 0.3171 | 0.4001 | 0.3215 | 0.3930 |
| $A_{2013}$ | 0.4534 | 0.3074 | 0.3857 | 0.3053 | 0.3879 |
| $A_{2014}$ | 0.4809 | 0.3375 | 0.3944 | 0.3578 | 0.3748 |
| $A_{2015}$ | 0.5959 | 0.7282 | 0.6890 | 0.7313 | 0.6866 |
| $A_{2016}$ | 0.5427 | 0.7690 | 0.7006 | 0.7597 | 0.7063 |
| $A_{2017}$ | 0.5674 | 0.7223 | 0.6946 | 0.7138 | 0.7004 |
| $A_{2018}$ | 0.5463 | 0.5918 | 0.5862 | 0.5980 | 0.5830 |
| $A_{2019}$ | 0.5340 | 0.6748 | 0.6501 | 0.6673 | 0.6592 |
| $A_{2020}$ | 0.5537 | 0.7057 | 0.6661 | 0.7113 | 0.6497 |
| $A_{2021}$ | 0.4357 | 0.5186 | 0.5143 | 0.5122 | 0.5214 |
| $A_{2022}$ | 0.3942 | 0.2110 | 0.2782 | 0.2119 | 0.2808 |

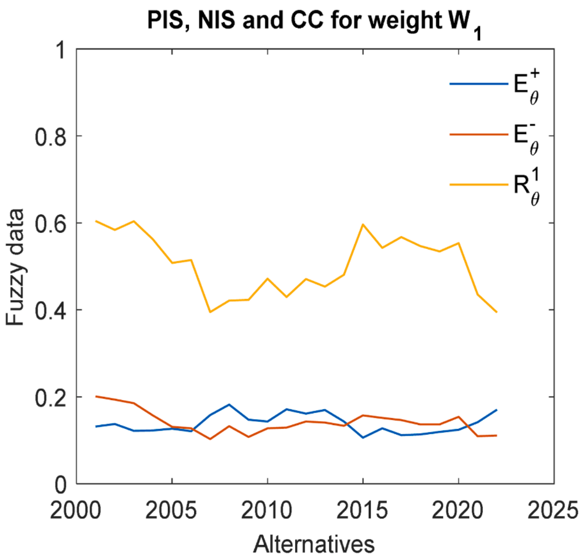



**Fig. 8.** First decision-maker ($DM^1$).

$I_G^- = (\widetilde{N}_{G1}^-, \widetilde{N}_{G2}^-, \cdots, \widetilde{N}_{GK}^-) = \left(\min_\theta \propto_1 R(A_\theta)^1, \min_\theta \propto_2 R(A_\theta)^2, \cdots, \min_\theta \propto_K R(A_\theta)^K\right).$

Step 5. Compute each alternative $A_\theta$, the distance from the group, $I_G^+$ and $I_G^-$ correspondingly as:

$D_{G\theta}^+ = \sqrt[2]{\sum_{k=1}^K (\propto_K R(A_\theta)^K - \widetilde{N}_{Gk}^+)^2}$ with $\theta = 1, 2, 3, \cdots, m$ and $D_{G\theta}^- = \sqrt[2]{\sum_{k=1}^K (\propto_K R(A_\theta)^K - \widetilde{N}_{Gk}^+)^2}$ with $\theta = 1, 2, 3, \cdots, m$

Step 6. Compute the group relative closeness $R_G(A_\theta)$ for each alternative $A_\theta$ with respect to ideal solution.

$R_G(A_\theta) = \frac{D_{G\theta}^-}{D_{G\theta}^+ + D_{G\theta}^-}$ , $\theta = 1, 2, 3, \cdots, m.$

From the above procedure, we conclude that the integration of type-1 fuzzy TOPSIS with regular TOPSIS, can improve the conventional MCGDM method and successfully handles decision matrix's uncertainty. It enhances distance computations using fuzzy Euclidean measures, refines group weighing mechanisms, and uses fuzzy logic to manage ambiguous preferences, in contrast to previous techniques. This leads to a more accurate, adaptable, and strong framework for group decision-making that more accurately captures consensus and subjective evaluations.

We show the method of subsequent application to a study involving a global crude oil price spill. The decision-matrix, which is comprised of alternatives, criteria, and the decision makers preferences expressed as degree of important weights, constructed using

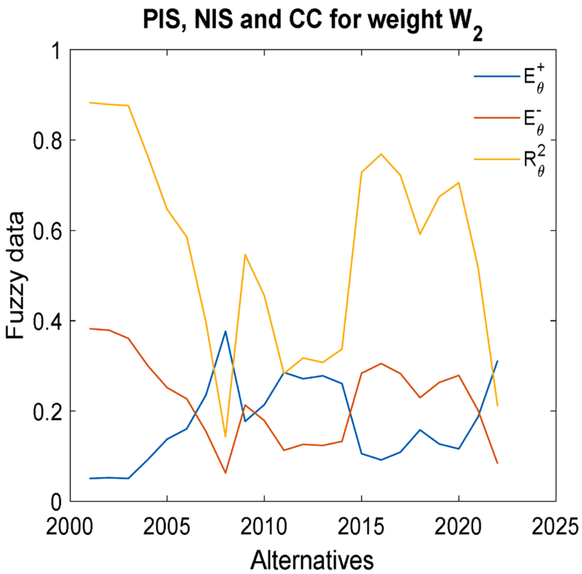



**Fig. 9.** Second decision-maker ($DM^2$).

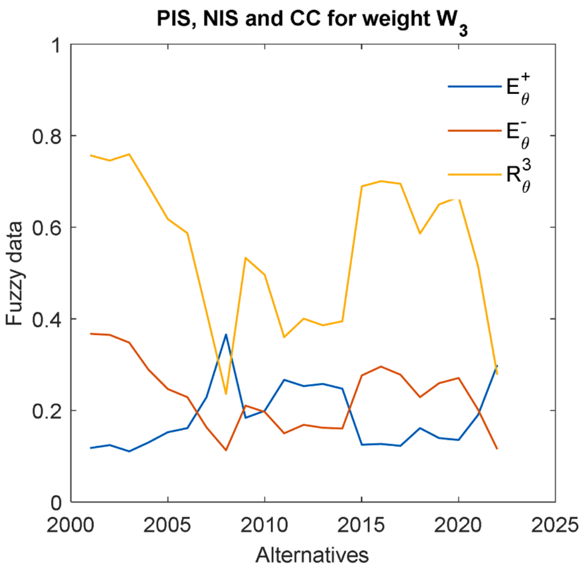



**Fig. 10.** Third decision-maker ($DM^3$).

the study description for the global oil price spill that occurred from the years 2001 to 2022.

## 4. Case study – crude oil price in world market

Crude oil is a naturally occurring liquid petroleum product made up of deposits of hydrocarbon and other organic compounds that were once living things and were existed millions of years ago. After being exposed to heat, pressure, and layers of sand, silt, and rock, these organisms finally transformed into a kind of fossil fuel that is refined into usable goods like gasoline, diesel, liquefied petroleum gases, and feedstock for the petrochemical sector, is presented in Fig. 4. Global commodity trading activity is centred around the WTI standard for US crude oil. In trading economics, the prices of crude oil are derived from financial instruments such as over-the-counter (OTC) and contract for difference (CFD). For buyers and sellers of crude oil such as WTI, OPEC Reference Basket, Brent crude, Tapis Crude, Dubai Crude, Bonny Light, Isthmus, Urals oil, and Western Canadian Select (WCS), the oil price, generally refers to the spot price of a barrel (roughly, 159 l) of benchmark crude oil. Instead of any nation's internal productions level, the price of oil is decided by global supply and demand [11].

Throughout the nineteenth and early twentieth centuries, the price of crude oil was largely stable worldwide. This changed in the

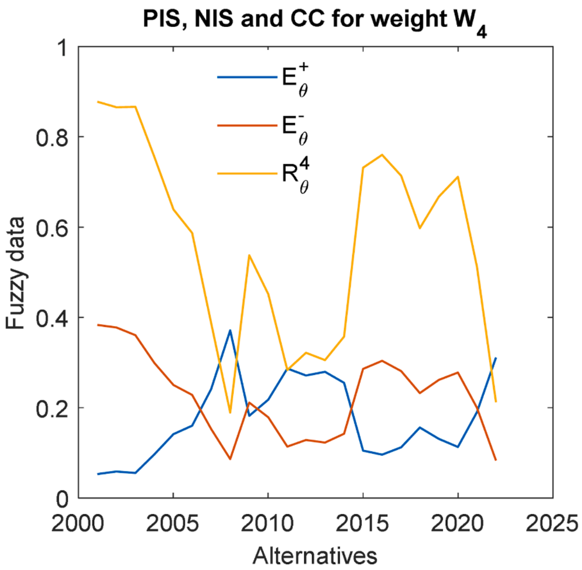



**Fig. 11.** Fourth decision-maker ($DM^4$).

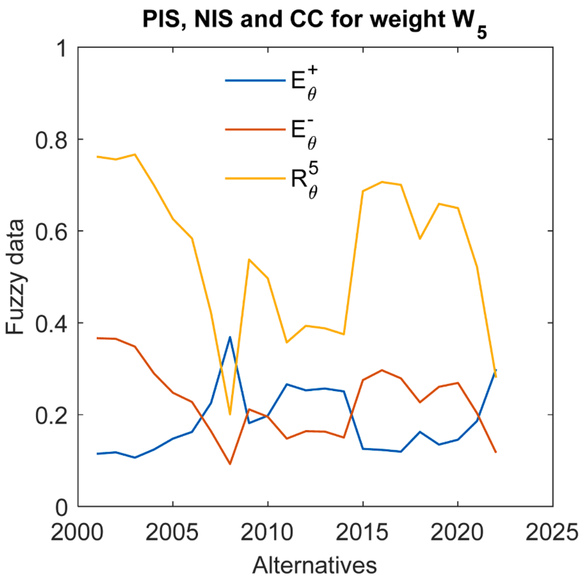



**Fig. 12.** Fifth Decision-maker ($DM^5$).

year 1970, when the price of oil increased significantly on a global scale [51]. In the past, stocks of the global economy's growth had an impact on oil prices, as have stocks related to supply, demand, and storage of oil [24]. The largest oil price declines in modern history from 2014 to 2016, as a result of the financial crisis of 2007–2008 and the more recent 2013 oil supply [26]. After its 40 years expert embargo was lifted, the United States resumed exporting oil in 2015, ranking third among the world's oil-producing nations [26,28].

When the COVID-19 epidemic first started in 2020, Russia-Saudi Arabia oil price war caused a 65 % drop in world oil prices. When the world began to recover from the COVID-19 recession, demand for energy surged globally in 2021, pushing record-high prices [1]. Global tight oil inventories by December 2021 as a result of an unanticipated recovery in the demand for oil from the US, China, and India, as well as demands to hold the line on spending from investors in the US shale industry. Concerns regarding the rising cost of gasoline which touched a record high in the United Kingdom on January18, 2022 were voiced as the price of Brent crude oil rose to its highest point since 2014- $88 [29].

A set volume of spilled crude oil ($159 \times 10^3 m^3$) was used for inquiry purpose. Modelling oil trajectories based on data on the kind of oil, spill point location, spilled volume, weather, and time of year has been used to study response tactics to oil spills [52]. In the various types of equipment, this form of study enables the simulation of many scenarios and some configurations of battle tactics. The

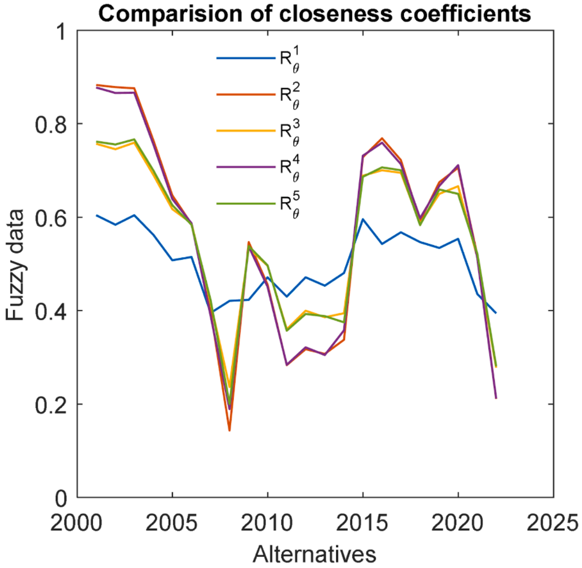



**Fig. 13.** Comparision analysis of CC for each DM.

**Table 6**
Application of type-1 fuzzy TOPSIS with first degree of importance weight $\alpha = (0.20, 0.20, 0.20, 0.20, 0.20)$ to the relative closeness matrix results in group positive ideal solutions and group negative ideal solutions.

| *Al.* \ *Cr.* | $(\alpha_1 R(A_\theta)^1, \alpha_2 R(A_\theta)^2, \alpha_3 R(A_\theta)^3, \alpha_4 R(A_\theta)^4, \alpha_5 R(A_\theta)^5)$ | $D^+_{G\theta}$ | $D^-_{G\theta}$ | $R_G(A_\theta)$ | Ranking order |
|---|---|---|---|---|---|
| $A_{2001}$ | (0.1209, 0.1766, 0.1514, 0.1756, 0.1523) | 0.0011 | 0.2570 | 0.9958 | 1 |
| $A_{2002}$ | (0.1168 ,0.1757, 0.1492, 0.1731, 0.1511) | 0.0060 | 0.2530 | 0.9769 | 3 |
| $A_{2003}$ | (0.1207, 0.1753, 0.1518, 0.1733, 0.1533) | 0.0026 | 0.2556 | 0.9898 | 2 |
| $A_{2004}$ | (0.1124, 0.1528, 0.1380, 0.1511, 0.1400) | 0.0401 | 0.2180 | 0.8445 | 5 |
| $A_{2005}$ | (0.1016, 0.1293, 0.1235, 0.1279, 0.1252) | 0.0805 | 0.1783 | 0.6890 | 10 |
| $A_{2006}$ | (0.1029, 0.1172, 0.1174, 0.1174, 0.1168) | 0.0988 | 0.1598 | 0.6179 | 12 |
| $A_{2007}$ | (0.0790, 0.0794, 0.0830, 0.0776, 0.0843) | 0.1741 | 0.0859 | 0.3304 | 16 |
| $A_{2008}$ | (0.0842, 0.0287, 0.0473, 0.0379, 0.0401) | 0.2568 | 0.0053 | 0.0204 | 22 |
| $A_{2009}$ | (0.0846, 0.1093, 0.1067, 0.1075, 0.1076) | 0.1209 | 0.1394 | 0.5355 | 13 |
| $A_{2010}$ | (0.0943, 0.0911, 0.0992, 0.0903, 0.0994) | 0.1448 | 0.1144 | 0.4413 | 15 |
| $A_{2011}$ | (0.0860, 0.0567, 0.0720, 0.0568, 0.0715) | 0.2069 | 0.0527 | 0.2031 | 20 |
| $A_{2012}$ | (0.0942, 0.0634, 0.0800, 0.0643, 0.0786) | 0.1915 | 0.0685 | 0.2635 | 18 |
| $A_{2013}$ | (0.0907, 0.0615, 0.0771, 0.0611, 0.0776) | 0.1965 | 0.0636 | 0.2446 | 19 |
| $A_{2014}$ | (0.0962, 0.0675, 0.0789, 0.0716, 0.0750) | 0.1866 | 0.0718 | 0.2778 | 17 |
| $A_{2015}$ | (0.1192, 0.1456, 0.1378, 0.1463, 0.1373) | 0.0477 | 0.2114 | 0.8159 | 6 |
| $A_{2016}$ | (0.1085, 0.1538, 0.1401, 0.1519, 0.1413) | 0.0389 | 0.2200 | 0.8496 | 4 |
| $A_{2017}$ | (0.1135, 0.1445, 0.1389, 0.1428, 0.1401) | 0.0501 | 0.2097 | 0.8072 | 7 |
| $A_{2018}$ | (0.1093, 0.1184, 0.1172, 0.1196, 0.1166) | 0.0960 | 0.1624 | 0.6286 | 11 |
| $A_{2019}$ | (0.1068, 0.1350, 0.1300, 0.1335, 0.1318) | 0.0682 | 0.1910 | 0.7369 | 9 |
| $A_{2020}$ | (0.1107, 0.1411, 0.1332, 0.1423, 0.1299) | 0.0580 | 0.2000 | 0.7752 | 8 |
| $A_{2021}$ | (0.0871, 0.1037, 0.1029, 0.1024, 0.1043) | 0.1289 | 0.1306 | 0.5034 | 14 |
| $A_{2022}$ | (0.0788, 0.0422, 0.0556, 0.0424, 0.0562) | 0.2372 | 0.0230 | 0.0884 | 21 |
| $I^+_G$ | (0.1209, 0.1766, 0.1518, 0.1756, 0.1533) | | | | |
| $I^-_G$ | (0.0788, 0.0287, 0.0473, 0.0379, 0.0401) | | | | |

information extracted from the spot trajectory prediction is crucial for developing a backup plan. In this scenario, utilising the computer software OILMAP package, the study has been simulated [53].

There are 22 choices outlined for this case study. We select these options based on the number of years, which ranges from 2001 to 2022. Based on the fluctuating pricing in each year, the set of alternatives is assessed based on each criterion, which can be regarded as typical in numerous circumstances. These factors included, but are not limited to the expense of cleaning, the price of oil that reaches the coast, and the oil composed by military evolution. They represent interests, putting particular focus on the pollution brought on by oil spills. Essentially, decision-makers, alternatives, and criteria make up the GDM process.

The criteria in this paper are the oil costs for each year at average various aspects, and the alternatives reflect the number of years. Three separate oil companies, an NGO, and Environmental Agency are the decision-makers. The simulation results for various containment and collecting strategies related to the combined dispersion of the spilled oil price are shown in Table 2. Other criteria are

**Table 7**
Application of type-1 fuzzy TOPSIS with second degree of importance weight $\alpha = (0.10,\ 0.10,\ 0.60,\ 0.10,\ 0.10)$ to the relative closeness matrix results in the group positive ideal solutions and group negative ideal solutions.

| *Al.* \ *Cr.* | $(\alpha_1 R(A_\theta))^1, \alpha_2 R(A_\theta))^2, \alpha_3 R(A_\theta))^3, \alpha_4 R(A_\theta))^4, \alpha_5 R(A_\theta))^5)$ | $D^+_{G\theta}$ | $D^-_{G\theta}$ | $R_G(A_\theta)$ | Ranking order |
|---|---|---|---|---|---|
| $A_{2001}$ | (0.0604, 0.0883, 0.4543, 0.0878, 0.0762) | 0.0013 | 0.3339 | 0.9962 | 1 |
| $A_{2002}$ | (0.0584, 0.0879, 0.4475, 0.0866, 0.0755) | 0.0084 | 0.3269 | 0.9748 | 3 |
| $A_{2003}$ | (0.0604, 0.0876, 0.4555, 0.0867, 0.0767) | 0.0013 | 0.3347 | 0.9961 | 2 |
| $A_{2004}$ | (0.0562,0.0764,0.4141,0.0755,0.0700) | 0.0455 | 0.2898 | 0.8643 | 5 |
| $A_{2005}$ | (0.0508, 0.0646, 0.3706, 0.0639, 0.0626) | 0.0929 | 0.2426 | 0.7231 | 10 |
| $A_{2006}$ | (0.0515, 0.0586, 0.3521, 0.0587, 0.0584) | 0.1133 | 0.2223 | 0.6625 | 12 |
| $A_{2007}$ | (0.0395, 0.0397, 0.2490, 0.0388, 0.0421) | 0.2215 | 0.1141 | 0.3401 | 16 |
| $A_{2008}$ | (0.0421, 0.0144, 0.1418, 0.0190, 0.0200) | 0.3350 | 0.0027 | 0.0079 | 22 |
| $A_{2009}$ | (0.0423, 0.0546, 0.3200, 0.0537, 0.0538) | 0.1467 | 0.1891 | 0.5631 | 13 |
| $A_{2010}$ | (0.0471, 0.0456, 0.2977, 0.0451, 0.0497) | 0.1716 | 0.1641 | 0.4888 | 15 |
| $A_{2011}$ | (0.0430, 0.0283, 0.2159, 0.0284, 0.0357) | 0.2579 | 0.0777 | 0.2315 | 20 |
| $A_{2012}$ | (0.0471, 0.0317, 0.2400, 0.0321, 0.0393) | 0.2330 | 0.1028 | 0.3061 | 17 |
| $A_{2013}$ | (0.0453, 0.0307, 0.2314, 0.0305, 0.0388) | 0.2418 | 0.0940 | 0.2798 | 19 |
| $A_{2014}$ | (0.0481, 0.0337, 0.2366, 0.0358, 0.0375) | 0.2351 | 0.1002 | 0.2988 | 18 |
| $A_{2015}$ | (0.0596, 0.0728, 0.4134, 0.0731, 0.0687) | 0.0479 | 0.2879 | 0.8574 | 7 |
| $A_{2016}$ | (0.0543, 0.0769, 0.4204, 0.0760, 0.0706) | 0.0397 | 0.2959 | 0.8816 | 4 |
| $A_{2017}$ | (0.0567, 0.0722, 0.4168, 0.0714, 0.0700) | 0.0457 | 0.2907 | 0.8642 | 6 |
| $A_{2018}$ | (0.0546, 0.0592, 0.3517, 0.0598, 0.0583) | 0.1130 | 0.2224 | 0.6630 | 11 |
| $A_{2019}$ | (0.0534, 0.0675, 0.3900, 0.0667, 0.0659) | 0.0730 | 0.2628 | 0.7826 | 9 |
| $A_{2020}$ | (0.0554, 0.0706, 0.3996, 0.0711, 0.0650) | 0.0623 | 0.2732 | 0.8144 | 8 |
| $A_{2021}$ | (0.0436, 0.0519, 0.3086, 0.0512, 0.0521) | 0.1585 | 0.1770 | 0.5275 | 14 |
| $A_{2022}$ | (0.0394, 0.0211, 0.1669, 0.0212, 0.0281) | 0.3083 | 0.0273 | 0.0814 | 21 |
| $I^+_G$ | (0.0604, 0.0883, 0.4555, 0.0878, 0.0767) | | | | |
| $I^-_G$ | (0.0394, 0.0144, 0.1418, 0.0190, 0.0200) | | | | |

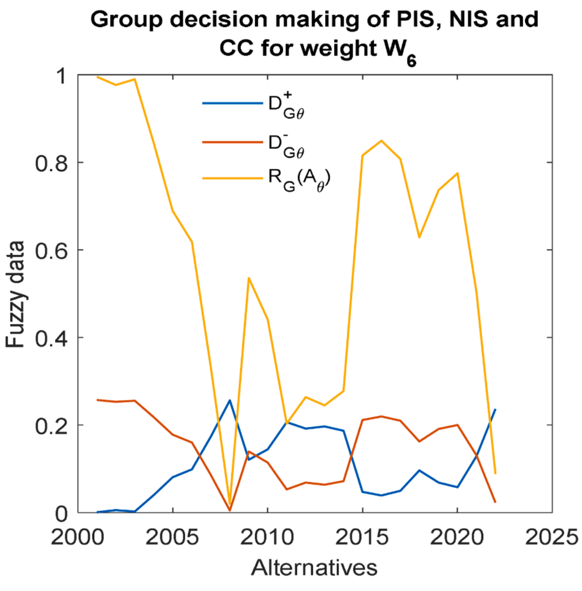



**Fig. 14.** First degree of importance weight.

employed, although the amount of oil at the cost (OC) per barrel in each year i.e. the years average closing price (YACP), years open price (YOP), years high price (YHP), years low price (YLP), and years closing price (YCP) are the best that has been adopted in this study. These are represented graphically in Figs. 5–7.

## 5. Results and discussion

In order to handle data uncertainties and take decision-makers preferences into account, the type-1 fuzzy TOPSIS method was created and applied, as demonstrated by the findings that are presented.

The decision matrix $D_M$, as outlined in Table 2, encompasses 22 alternatives and five criteria. In accordance with the type-1 fuzzy TOPSIS notation, the first criteria $C_{YACP}$ for the years YACP, the second criteria $C_{YOP}$ for YOP, the third criteria $C_{YHP}$ for YHP, the fourth

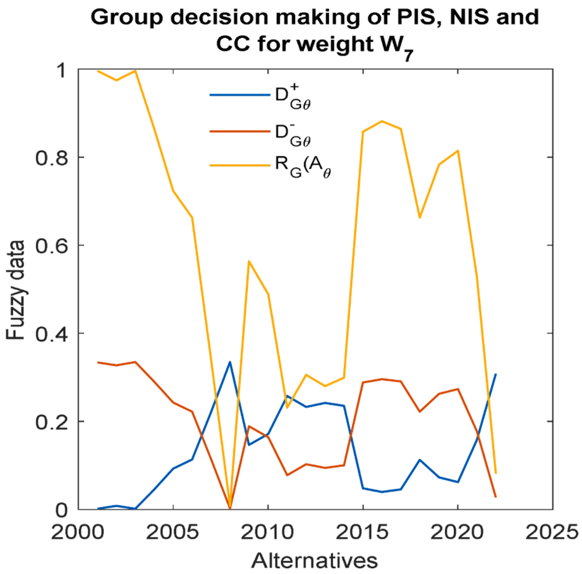



**Fig. 15.** Second degree of importance weight.

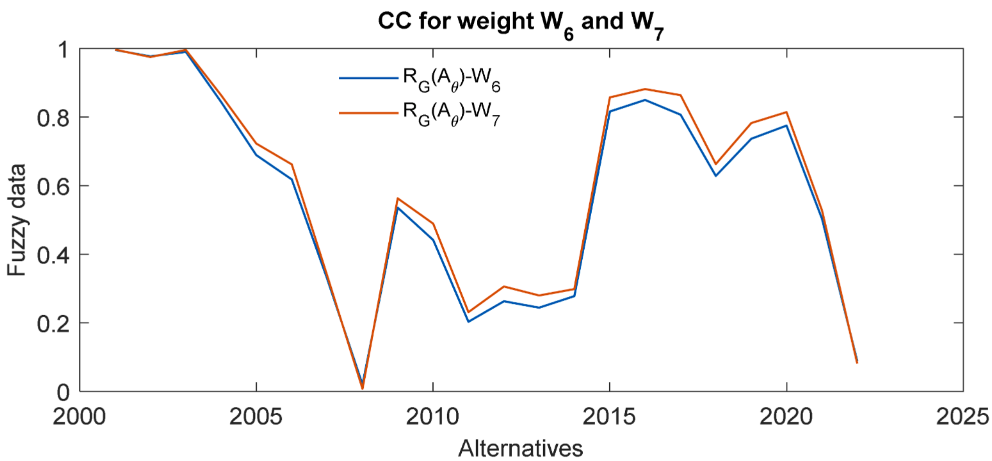



**Fig. 16.** Analysis of closeness values in two different weights.

criteria $C_{YLP}$ for the YLP, and fifth criteria $C_{YCP}$ for the YCP. Initially, the DM $D_M$ has been normalized. In this paper, we study only the positive values and the average value for all the criteria is 159 $\times 10^3$, normalized the data of the DM corresponding to:

$N_{\theta\sigma} = \frac{x_{\theta\sigma}}{MAX\widetilde{x_\sigma}}$, with $\theta = 1, 2, \cdots, m$; $\sigma = 1, 2, \cdots, n$.where $\widetilde{x}_\sigma$ gives the largest value for each criterion $C_\sigma$.

The data from our simulations, as presented in Table 2, is subjected to uncertainty due to various factors influencing the simulation of oil spills. These variables include among other, the location of the spill, the kind and volume of oil that was released, and the shifting weather conditions. The dynamic character of the maritime environment, which is marked by variables that changes over time, makes solving this problem especially difficult. We include a 10 % uncertainty in our study, which has an impact on the rating given to each possibility. Table 3 provides a through representation of the fuzzy DM, which is obtained through post-normalisation using triangular fuzzy numbers.

We introduce a weight vector $W$ to represent the weight assigned to each criterion according to the decision maker's viewpoint in order to account for this variance in priority. Each criterion has five levels of importance: very important, important, moderate, unimportant, and less important. Notably, three separate oil companies, an NGO, and the Environmental Agency are the five DM bodies involved in our study.

The weights assigned to the labels are 0.50, 0.20, 0.15, 0.10, and 0.05 respectively, as the sum of the weighted value of each criterion is one. Consequently, the weight vectors for $DM^1$ is $W_1 = (0.20, 0.20, 0.20, 0.20, 0.20)$, for $DM^2$ is $W_2 = (0.20, 0.05, 0.50, 0.10, 0.15)$, for $DM^3$ is $W_3 = (0.05, 0.20, 0.50, 0.15, 0.10)$, for $DM^4$ is $W_4 = (0.15, 0.10, 0.50, 0.05, 0.20)$, and for $DM^5$ is $W_5 = (0.10, 0.15, 0.50, 0.20, 0.05)$. Similarly, a weight vector corresponding importance $\alpha = (\alpha_1, \alpha_2, \alpha_3, \alpha_4, \alpha_5)$ is allocated to each decision makers. We report on the type-1 fuzzy TOPSIS application outcome for the decision maker in Table 4 for $DM^1$. Similarly, In Table 5, we presented all outcome values of $DM^2$, $DM^3$, $DM^4$, and $DM^5$ respectively. The graphical representation of the results of fuzzy data i.e. PIS, NIS and closeness coefficient values for each decision maker of different weights are shown in Figs. 8–12, respectively.

We create the relative closeness matrix using the relative closeness values, of $R_\theta^\sigma$, that are obtained after applying type-1 fuzzy TOPSIS to each DM. This matrix is displayed in Table 5. Fig. 13 provides the graphical representations of these closeness values of each decision maker. To represent equal importance across all DMs, we first select weight vector $\alpha = (0.10,\ 0.10,\ 0.60,\ 0.10,\ 0.10)$. Table 6 presents the normalized weighted closeness matrix, which is produced by multiplying the relative values of Table 5 by the weights.

Currently, we examine an alternative weight vector, $\alpha = (0.10,\ 0.10,\ 0.60,\ 0.10,\ 0.10)$. The normalised weighted closeness matrix Table 7 is produced by multiplying the relative closeness matrix Table 5 by these degree of importance weights. Thus, Figs. 14 and 15 shows the closeness coefficient, PIS, and NIS for the first and second sets of degree of important weights in both the line graph and narrative formats. And from the Tables 6 and 7 and Fig. 16 we conclude that the order of alternative is equal but distinct in two different cases of alternative i.e. $A_{2012} = A_{2014}$ and $A_{2014} = A_{2012}$ in both the degree of importance weighted vectors. Thus, the ranking order in the first degree of importance weight is

$A_{2001} > A_{2003} > A_{2002} > A_{2016} > A_{2004} > A_{2015} > A_{2017} > A_{2020} > A_{2019} > A_{2005} > A_{2018} > A_{2006} > A_{2009} > A_{2021} > A_{2010} > A_{2007} > A_{2014} > A_{2012} > A_{2013} > A_{2011} > A_{2022} > A_{2008}$ and the second degree of importance weight is

$A_{2001} > A_{2003} > A_{2002} > A_{2016} > A_{2004} > A_{2015} > A_{2017} > A_{2020} > A_{2019} > A_{2005} > A_{2018} > A_{2006} > A_{2009} > A_{2021} > A_{2010} > A_{2007} > A_{2012} > A_{2014} > A_{2013} > A_{2011} > A_{2022} > A_{2008}$.

Regarding ranking consistency and sensitivity to uncertainty, the comparative study between Krohling and Campanharo's technique [10] and the suggested type-1 FTOPSIS method provides significant insights. Up to ten choices are considered in their process, and the ranking order of the alternatives stays the same. Thus, the ranking order of 10 alternatives in first and second degree of importance weights are same [10], as:

$A_8 > A_9 > A_5 > A_{10} > A_7 > A_6 > A_4 > A_1 > A_2 > A_3$

From the ranking of first and second alternatives, found $A_8$ is the optimal solution [10]. When the number of alternatives increases more than 10 like 22, however, disparities appear and certain alternatives are ranked differently. Out of these different ranking order, we found $A_{2001}$ is the best alternative and $A_{2008}$ is least alternative for oil price of first and second degree of importance weights. But the ranking of first degree of importance weights and second degree of importance weights alternatives $A_{2012}$ and $A_{2014}$ are slightly different. This shows that the ranking behaviour gets more dynamic as the number of choices rises, maybe as a result of the increased complexity of decision-making in bigger datasets.

Furthermore, the ultimate ranking order is greatly influenced by the degree of important weight vector selection. The method's sensitivity to preference selection is shown by the fact that different weight assignments produce different values and changes in the ranking of alternatives. This suggests that in order to guarantee accurate and significant findings, decision-makers need to carefully choose weight vectors. The inclusion of ambiguity in the decision-making process is another essential component.

In order to demonstrate that uncertainty may influence ultimate judgments, the study applies a 10 % uncertainty factor to the DM data, which influences the ranking of options. The degree of uncertainty in decision-making might vary depending on the application and the number of options taken into account. This highlights how crucial strong uncertainty modelling is too fuzzy MCDM techniques in order to improve decision stability and dependability at various problem scales.

By combining type-1 fuzzy TOPSIS with normal TOPSIS, the suggested method improves on conventional MCGDM techniques and successfully handles decision matrix's uncertainty. In contrast to current techniques, it enhances distance computations using fuzzy Euclidean measures, refines group weighting processes, and uses fuzzy logic to manage ambiguous preferences. This leads to a framework for collective decision-making that is more accurate, adaptable, and strong and that more accurately reflects consensus and subjective assessments.

## 6. Conclusions and future works

Fuzzy decision-matrix problems are common application domain for MCDM in the real world. This paper is devoted for improving and expanding the TOPSIS and the type-1 fuzzy TOPSIS approaches. We provide a novel type-1 fuzzy TOPSIS technique that is especially suited for GDM. It is intended to handle uncertain multi-criteria choice problems that take decision-makers preference into account. The developed method facilitates the identification of optimal alternatives, contributing to effective DM. In order to demonstrate the suitability and efficacy of our suggested technique, we carried out a case study employing the world's biggest reserves of crude oil.

The limitation of the study, we applied the method to evaluate twenty-two different alternatives based on five criteria, involving the perspectives of five DMs for determining the best options when it comes to the global oil price. The study relies on type-1 fuzzy sets, which might not adequately reflect uncertainty, and has limited validation across a variety of domains. It may also have expandability problems with big decision matrices. Results may also be less reliable due to the subjectivity of decision-makers' preferences and the lack of thorough comparisons with other MCDM techniques.

In future research, the method can be applied to other MCDM problems with a finite number of alternatives (>22), criteria (more than 5), and decision makers (more than 5). Future studies might investigate sensitivity analysis, AI-based decision support systems, type-2 or neutrosophic fuzzy TOPSIS, hybridization with other MCDM techniques like AHP, DEMATEL, etc. and wider real-world applications to improve robustness and usefulness. The robustness and adaptability of the proposed method makes it a valuable asset in addressing diverse DM scenarios beyond the context of the present study.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The entire work has been carried by our own interest and mutual understanding with equal contribution.

## Data availability

Data will be made available on request.

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